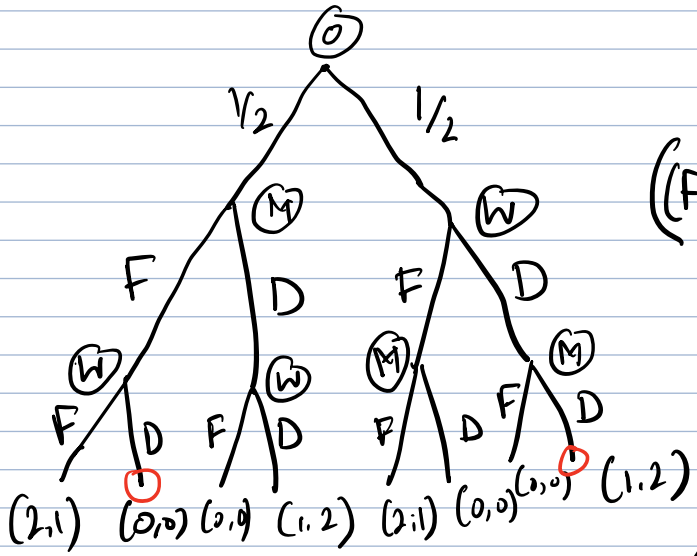
Class 20

$\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N \cup \{0\}}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$



$((F, F, D), (D, F, D))$

$u_M((F, F, D), (D, F, D))$

$= \frac{1}{2} \times 0 + \frac{1}{2} \times 1 = 1/2$

$u_W((F, F, D), (D, F, D))$

$= \frac{1}{2} \times 0 + \frac{1}{2} \times 2 = 1$

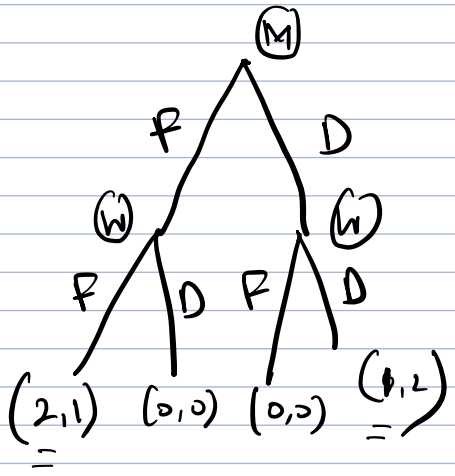
Nash Equilibrium A strategy profile s^* is an NE if for all $i \in N$, $s_i \in S_i$

$$u_i(s^*) \geq u_i(s_i, s^*_{-i})$$

$S_1, \ldots, S_n$

$\langle N, (S_i)_{i \in N}, (\hat{u}_i)_{i \in N} \rangle$ is corresponding strategic form game.

R-1 For an extensive form game Γ, a strategy profile s^* is an NE iff s^* is an NE of corresponding strategic form game.



$(F, (D, F))$ — not an NE, W has a +ve deviation

$(F, (F, F))$ — an NE / not SPE

$(F, (F, D))$ — an NE / SPE

$(D, (D, D))$ — an NE / not SPE

$(D, (F, D))$ — is not an NE; M has a +ve deviation

	(F,F)	(F,D)	(D,F)	(D,D)
F	(2,1)	(2,1)	(0,0)	(0,0)
D	(0,0)	(1,2)	(0,0)	(1,2)

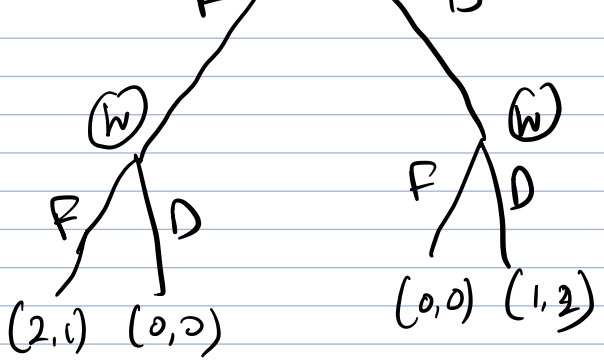
Subgame perfect Equilibrium

A strategy profile s^* is called a subgame perfect Equilibrium (SPE) if, for every subgame of the game, s^* restricted to that subgame is an NE of that subgame.

Every SPE is also an NE of the game.

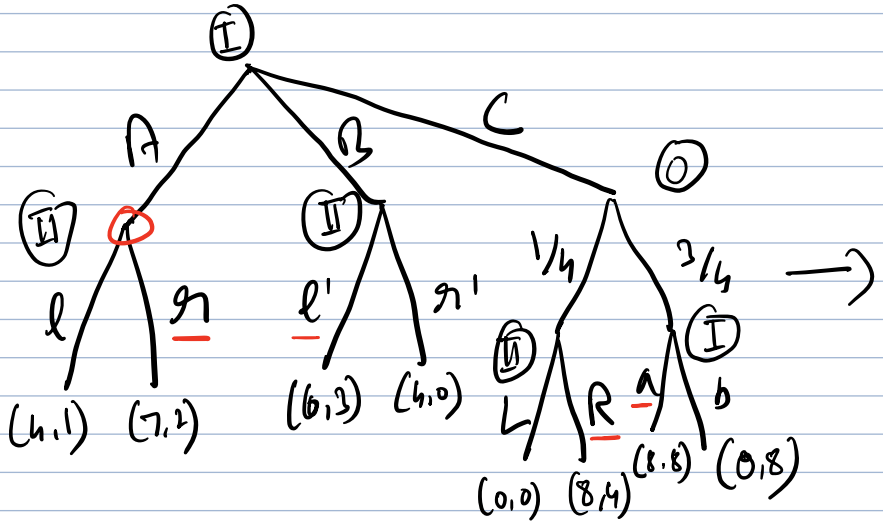
Backward Induction Equilibrium

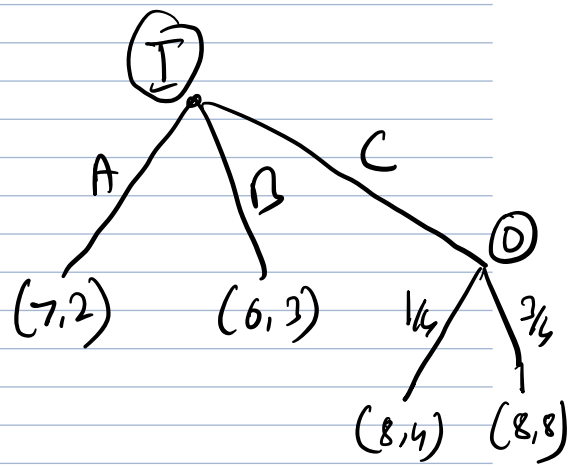


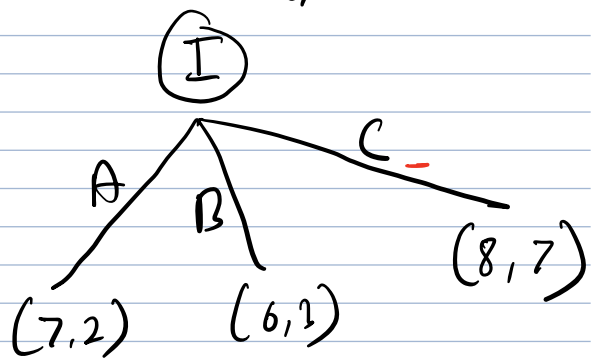




(F, (F, D))







((C, a), (r, ℓ', R))

is the backward induction Equilibrium.